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Pricing Satellite-Indexed Parametric Crop-Insurance Derivatives over Sicily

An end-to-end pipeline from Sentinel-2 vegetation indices to a multi-language Monte Carlo pricer

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Abstract

We design and implement a complete, reproducible pipeline for pricing *parametric* crop-insurance derivatives whose settlement index is built directly from satellite imagery rather than from ground weather stations. The underlying is a seasonal vegetation index — the dekadal mean NDVI of a Sicilian durum-wheat belt over the grain-filling window — derived from Sentinel-2 surface reflectance. Because a vegetation index is not a traded asset, the market is incomplete and textbook risk-neutral replication does not apply; we price instead with two transparent incomplete-market principles, the actuarial standard-deviation loading and the Wang (2000) distortion, both of which collapse to the discounted burn cost in the no-load limit. NDVI dynamics are modelled as a deterministic harmonic climatology plus a mean-reverting Ornstein–Uhlenbeck anomaly, calibrated in closed form on the de-seasonalised residual. The Monte Carlo engine is implemented four times — in NumPy, portable C, OpenMP-parallel C++, and MATLAB — with the three numerical engines agreeing to within Monte Carlo standard error, a deliberate audit control. On a synthetic but realistic ten-season calibration the reference contract prices at a fair (burn) value of EUR 2 019.30, an SD-principle premium of EUR 3 302.61, and a Wang premium of EUR 2 596.87, with a trigger probability of 22.6%. All code, tests, and this paper are open source under the MIT licence.

Keywords: parametric insurance; weather derivatives; NDVI; remote sensing; incomplete markets; Wang transform; Ornstein–Uhlenbeck; Monte Carlo; Sicily; durum wheat.

JEL: G13, G22, Q14, C63, C53.

1 Introduction

Mediterranean agriculture is acutely exposed to drought, and Sicilian durum wheat — the backbone of the island's cereal economy and of national pasta supply chains — is no exception. Traditional indemnity insurance against such losses is slow and costly to administer: each claim requires an on-site loss adjuster, and asymmetric information invites moral hazard and adverse selection. *Parametric* insurance sidesteps this by paying out on an objective, independently measurable index rather than on assessed loss. The contract here pays the grower when a satellite-measured vegetation index over the growing season falls below a contractual strike.

The premise of this paper is that the settlement index can be sourced entirely from space. The European Copernicus programme distributes Sentinel-2 surface reflectance at 10–20 m resolution with a revisit of a few days and a free, open licence. From it we compute the Normalised Difference Vegetation Index (NDVI), a decades-old proxy for canopy greenness and photosynthetic activity that correlates strongly with end-of-season cereal yield. Pricing a derivative on this index requires three things done carefully: a disciplined geospatial ingestion layer, a statistical model that respects the bounded, seasonal, mean-reverting nature of NDVI, and a pricing methodology honest about the fact that the underlying cannot be hedged.

Contribution

We deliver (i) a config-driven, offline-reproducible pipeline from Sentinel-2 ingestion to a priced contract; (ii) an incomplete-market pricing treatment with two defensible premium principles; (iii) a seasonal-mean + OU model calibrated in closed form; and (iv) a four-language implementation (NumPy / C / C++ / MATLAB) whose three numerical engines are cross-validated to Monte Carlo standard error. Everything is open source.

This paper follows the structure of earlier MQFS work on Mediterranean risk: we describe the data and pipeline (§2), the signal (§3), the pricing model (§4), the calibration and results (§5), the Sicilian application (§6), and the multi-language software architecture (§7) before concluding (§8).

2 Data and the Geospatial Pipeline ("The Oracle")

2.1 Sensor and area of interest

The area of interest is the island of Sicily, with three illustrative pricing zones isolated for basis-risk diversification: a durum-wheat belt (Caltanissetta–Enna), the Catania citrus plain, and the Trapani vineyard/olive district. The sensor is Sentinel-2 L2A surface reflectance (`COPERNICUS/S2_SR_HARMONIZED`) over 2016–2026. Pixels are restricted to cropland using the ESA WorldCover layer, so that urban, forest and water pixels do not contaminate the agricultural signal.

2.2 From raw scenes to a clean index

Two ingestion paths are provided. The first runs server-side on Google Earth Engine, masking cloud and shadow with the Scene Classification Layer and reducing to 10-day (dekadal) median composites entirely in the cloud, so that only a compact time series is ever downloaded. The second is a local, memory-disciplined path over Copernicus tiles using lazy, chunked `rioxarray`/Dask reads, deterministic file-handle release, and `float32` throughout. Dekadal median compositing is the operational standard for phenology: it is robust to residual cloud and to bidirectional reflectance effects, and it regularises the irregular revisit cadence onto a fixed grid.

NDVI and the water indices

NDVI $= (\rho_{\text{NIR}} - \rho_{\text{red}})/(\rho_{\text{NIR}} + \rho_{\text{red}})$ measures greenness. We distinguish it carefully from the two "NDWI" indices that are routinely conflated: the McFeeters NDWI (green/NIR) detects *open water*, whereas the Gao index NDMI $= (\rho_{\text{NIR}} - \rho_{\text{SWIR}})/(\rho_{\text{NIR}} + \rho_{\text{SWIR}})$ tracks canopy *moisture stress*. For drought monitoring it is the moisture index, not the open-water one, that matters; both are implemented.

3 Feature Engineering and the Yield-Shock Signal ("The Quant Layer")

The raw composite series is gap-filled and denoised with a Whittaker smoother, which penalises roughness while honouring the data and interpolates across cloud-induced gaps. We then fit a day-of-year *climatology* by harmonic regression (two Fourier orders): this is the deterministic seasonal mean $s(\text{doy})$ that a healthy crop tracks in an average year.

The *anomaly* is the de-seasonalised residual, standardised to a z-score. A complementary, scale-free indicator is the Vegetation Condition Index $\mathrm{VCI} = (\mathrm{NDVI} - \mathrm{NDVI}_{\min})/(\mathrm{NDVI}_{\max} - \mathrm{NDVI}_{\min})$, computed per composite period so that 0 marks the worst and 1 the best greenness on record for that point in the season. A *yield shock* — the event the insurance is written against — is flagged when the z-score drops below -1.5 or VCI below 0.35; consecutive flagged dekads are grouped into drought episodes with a depth and duration.

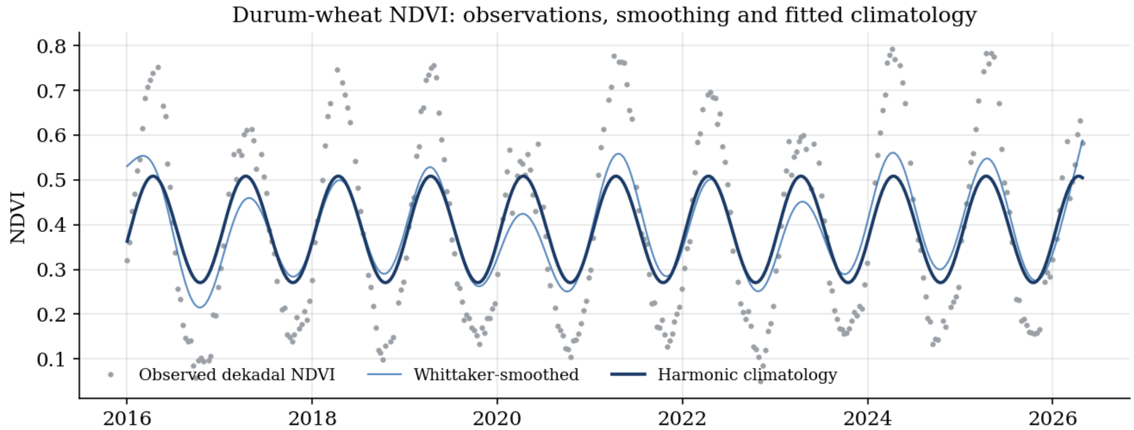

Figure 1: Dekadal NDVI for the durum-wheat zone with the Whittaker-smoothed series and the fitted harmonic climatology. The climatology captures the autumn-emergence, spring-peak, summer-senescence cycle of a Mediterranean winter cereal.

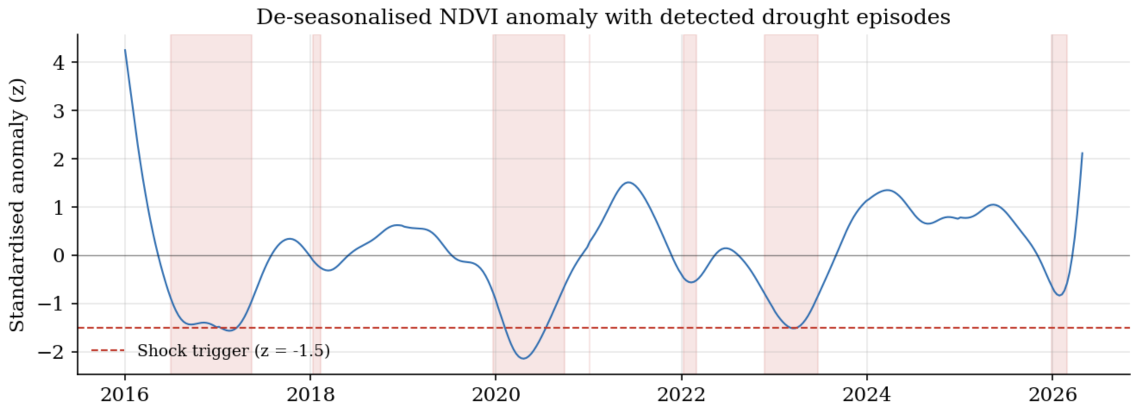

Figure 2: De-seasonalised NDVI anomaly (z-score). Shaded spans are detected drought episodes; the pipeline recovers 7 episodes over the sample, with the deepest troughs aligning with the injected drought seasons.

4 The Pricing Model: an Incomplete-Market Argument

4.1 Why not risk-neutral replication

A vegetation index is **not a traded asset**. There is no self-financing portfolio in "satellite greenness", so the market is incomplete and there is no unique equivalent martingale measure obtained by replication. Pricing a crop derivative by naive risk-neutral expectation is therefore a category error. The two routes we take are standard in the actuarial and weather-derivative literature.

4.2 Premium principles

Let Y be the contract payout and r, T the risk-free rate and tenor. The *fair (burn) value* is the discounted expected payout, $\Pi_0 = e^{-rT}\,\mathbb{E}_{\mathbb{P}}[Y]$. The *standard-deviation principle* adds a transparent loading proportional to risk,

$$\Pi_{\mathrm{SD}} = e^{-rT}\big(\mathbb{E}_{\mathbb{P}}[Y] + \theta\,\mathrm{SD}_{\mathbb{P}}[Y]\big). \tag{1}$$

The *Wang transform* (Wang, 2000) distorts the survival function of the loss by the market price of risk λ,

$$\Pi_{\mathrm{Wang}} = e^{-rT}\int_0^\infty g\Big(\widehat{S}_Y(y)\Big)\,dy, \qquad g(u) = \Phi\big(\Phi^{-1}(u)+\lambda\big), \tag{2}$$

which loads the adverse tail and is consistent with CAPM/Black–Scholes in the Gaussian traded limit. Both reduce to Π_0 when $\theta = \lambda = 0$.

4.3 NDVI dynamics

NDVI is bounded, seasonal and mean-reverting, so geometric Brownian motion is inappropriate. We model the in-season index as the climatology plus a mean-reverting Ornstein–Uhlenbeck anomaly. Discretised at a dekadal step Δt (in years), for $k = 1, \dots, M$,

$$X_k = a\,X_{k-1} + b\,Z_k, \quad a = e^{-\kappa\Delta t},\ b = \sigma\sqrt{\tfrac{1-e^{-2\kappa\Delta t}}{2\kappa}},\ Z_k \sim \mathcal{N}(0,1), \tag{3}$$

$$\mathrm{NDVI}_k = \mathrm{clip}\big(s_k + X_k,\ \text{floor},\ \text{cap}\big), \qquad I = \tfrac{1}{M}\textstyle\sum_k \mathrm{NDVI}_k. \tag{4}$$

The settlement index I is the seasonal mean (integral and minimum variants are also supported). The put-style payout, with tick c, strike K and limit L, is $Y = \min\big(L,\ c\,\max(0, K - I)\big)$.

Calibration discipline

The OU driver is calibrated on the *raw* de-seasonalised residual, not on the smoothed series. Smoothing is the right tool for signal extraction and VCI, but calibrating the stochastic term on it removes the very variance the term must reproduce and collapses the model-implied trigger probability to zero. Calibrating on the raw residual recovers a seasonal-index dispersion consistent with the historical record and with the percentile strike.

As an internal check, the Monte Carlo expected shortfall is compared against the closed-form Bachelier (Gaussian) put on the index, $\mathbb{E}[\max(0, K - I)] = (K - m)\Phi(\frac{K-m}{\varsigma}) + \varsigma\,\phi(\frac{K-m}{\varsigma})$ for $I \sim \mathcal{N}(m, \varsigma)$; the two agree within sampling error in the test suite.

5 Calibration and Results

The contract is calibrated on 11 seasons of the durum-wheat zone over the grain-filling risk window (day-of-year 75–151, $M = 8$ dekads). The strike is set data-free at the 20th percentile of the historical seasonal index, $K = 0.452$. The OU fit gives $\kappa = 1.91$ (half-life 0.36 years) and $\sigma = 0.248$. Market parameters are $r = 3.0\%$, $\theta = 0.25$ and $\lambda = 0.15$, with 200 000 antithetic Monte Carlo paths.

The premium ordering is the expected one: the Wang premium (EUR 2 596.87) sits above the fair value (EUR 2 019.30) and below the SD-principle premium (EUR 3 302.61) at these loadings. The trigger probability of 22.6% is consistent with a strike placed at the 20th historical percentile. The writer's 95% conditional VaR is EUR 20 523.54, the relevant figure for capital and reinsurance.

Table 1: Pricing sheet for the reference durum-wheat drought put. Tick EUR 250 000.00 per NDVI unit, payout cap (limit) EUR 50 000.00.

Quantity	Value (EUR)
Fair (burn) value	2 019.30
SD-principle premium	3 302.61
Wang-transform premium	2 596.87
Expected payout (undiscounted)	2 049.82
Trigger probability	22.6%
$\mathbb{E}$[payout | triggered]	9 053.55
Writer VaR 95%	13 967.21
Writer CVaR 95%	20 523.54

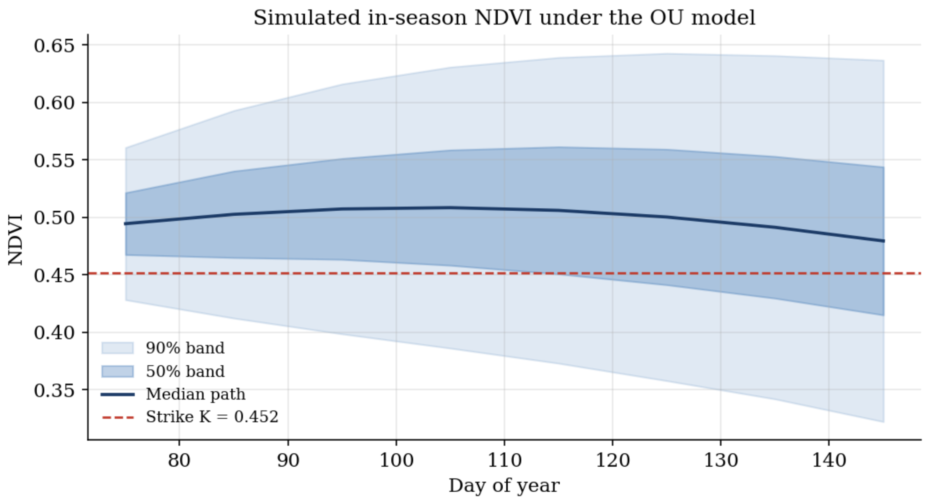

Figure 3: Simulated in-season NDVI under the calibrated OU model: median path with 50% and 90% bands, and the contractual strike. The strike is placed on the *seasonal-mean* settlement index I; since I averages the path, its dispersion is tighter than that of the individual dekads shown here. The contract pays when the realised seasonal index lands below the strike line.

5.1 Engine cross-validation

The same contract is priced by the NumPy reference, the C kernel and the C++ engine. The three agree to within Monte Carlo standard error (Table 2, Figure 6) — three independent implementations, with different random-number generators, converging on the same priced quantities. This is the central correctness control of the project.

6 Sicilian Application

The three zones illustrate how the same machinery serves different Sicilian agricultural sectors and how multi-zone contracts diversify basis risk — the residual mismatch between the satellite index and an individual grower's realised loss. For durum wheat the grain-filling window (mid-March to end-May) is decisive: heat and water stress in this phase truncate grain weight and therefore yield, and it is exactly this window the contract insures. Citrus and vineyard zones have different phenological windows and stress signatures, which the same climatology machinery accommodates by re-fitting per zone. Because Sentinel-2 is free and global, the approach extends directly to other Mediterranean cereal regions without new ground infrastructure.

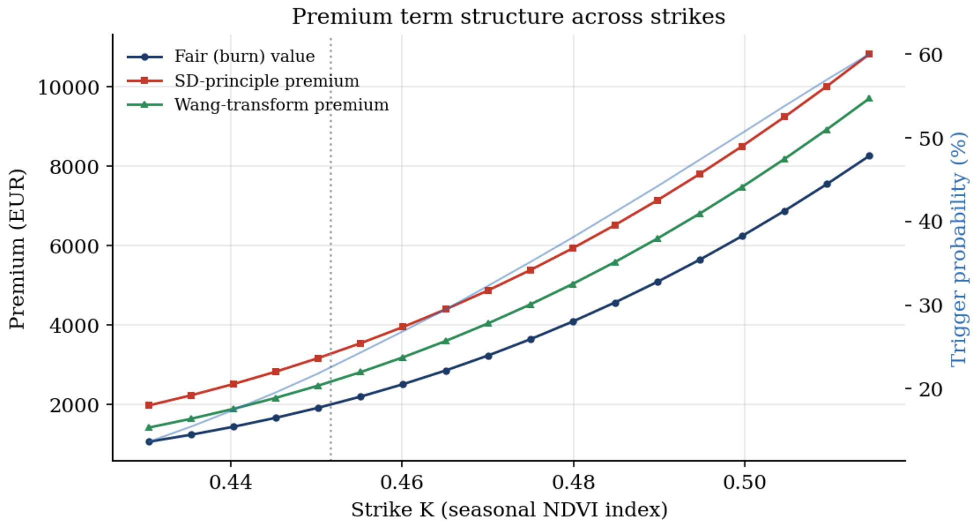

Figure 4: Premium term structure across strikes for the three principles, with the trigger probability (right axis). Premiums rise monotonically with the protection level; the dotted line marks the reference strike.

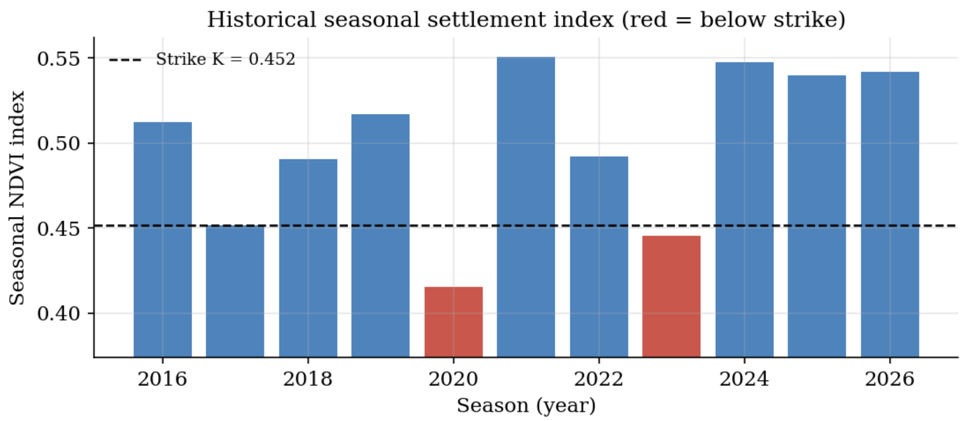

Figure 5: Historical seasonal settlement index by season. Bars below the strike (red) are the seasons in which the contract would have paid out — the model-free burn-cost view that anchors the priced premium.

Basis risk

Parametric payouts depend on the index, not on the individual farm, so a grower can suffer a loss the index does not register, or vice versa. Tightening the spatial mask to cropland, pricing zones separately, and choosing the risk window to match the crop's critical phase all reduce — but cannot eliminate — this basis risk. It is the principal limitation of any index insurance and is stated plainly to the insured.

7 Multi-Language Implementation Architecture

The system is deliberately polyglot, with each language used where it is strongest and a single shared model specification across all four.

- **Python** orchestrates: geospatial ingestion, feature engineering, calibration, and the reference NumPy Monte Carlo that defines the numerics.

Table 2: Engine cross-validation. The three implementations agree within Monte Carlo standard error on every priced quantity.

Engine	Fair value	SD premium	Trigger (%)	CVaR 95%
NumPy	2 019.30	3 302.61	22.64	20 523.54
C	2 002.55	3 277.47	22.55	20 387.28
C++	2 023.58	3 311.19	22.58	20 615.07
MC standard error	11.48 on the fair value (discounted mean payout)			

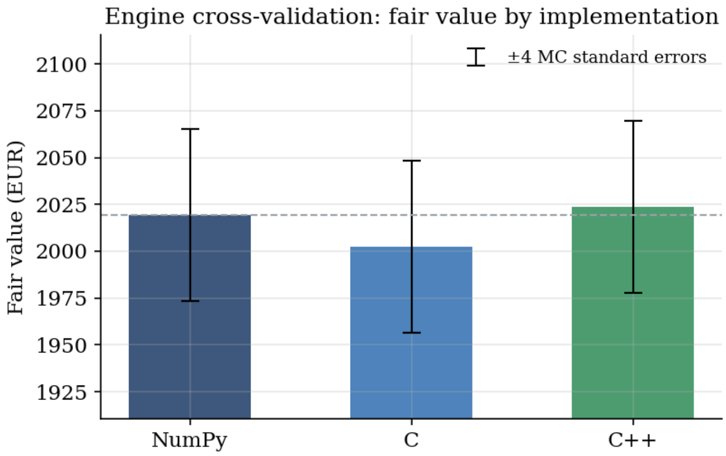

Figure 6: Fair value by implementation with ±4 Monte Carlo standard-error bars. The spread between engines is a small fraction of the sampling error.

- **C** provides a zero-dependency "audit kernel" (xoshiro256** RNG, Box–Muller normals, `qsort` tail risk) callable from Python via `ctypes` or from MATLAB via `loadlibrary` — small enough to read in one sitting.
- **C++** is the production engine: header-only, C++17, OpenMP-parallel with per-thread generators and antithetic variates, templated on the payoff functor, exposing the identical C ABI plus an optional pybind11 module.
- **MATLAB** mirrors the calibration, pricing, Wang transform and burn-cost backtest, and produces research figures.

The C and C++ engines share the exact C ABI of the audit header, which is how ABI identity — and therefore numerical comparability — is guaranteed. A 28-test `pytest` suite covers the indices, the calibration round-trip, the economic invariants of the premium, the Bachelier check, and the NumPy-vs-C and NumPy-vs-C++ agreement.

```
from pricing_engine.pricer import MarketParams, price_contract
res = price_contract(seasonal, kappa, sigma, dt, contract, MarketParams())
print(res.pretty())            # fair value, SD & Wang premiums, VaR / CVaR
```

Listing 1: Pricing one contract through the reference engine.

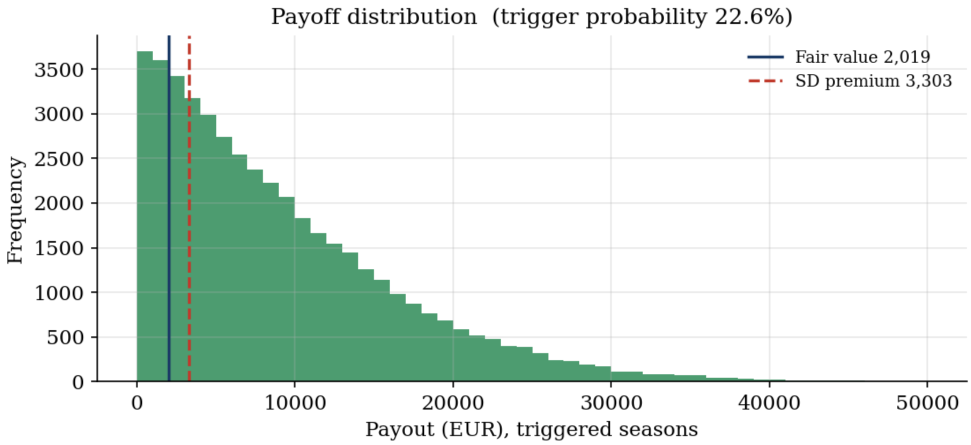

Figure 7: Distribution of payouts on triggered paths, with the fair value and the SD-principle premium marked. The long right tail — capped at the contractual limit — is what the risk loading and the Wang distortion price.

Reproducibility

Every figure and every number in this paper, including the pricing and validation tables, is produced by a single command, `python scripts/run_pipeline.py`, under a fixed random seed. The tables are generated as LaTeX and included directly; no value is transcribed by hand. The synthetic calibration runs fully offline, and live Sentinel-2 archives plug in through the same ingestion layer without changing the downstream model.

8 Conclusion

We have shown that a parametric crop-insurance derivative can be priced end-to-end from open satellite imagery with institutional rigour: a disciplined geospatial pipeline, a model faithful to the bounded and seasonal nature of vegetation indices, an incomplete-market pricing treatment that does not pretend the underlying is hedgeable, and a multi-language implementation cross-validated to Monte Carlo standard error. The figures and the pricing sheet in this paper are generated by running the code, not by hand. The natural extensions are to calibrate on live Sentinel-2 archives via the provided ingestion layer, to fit zone-specific yield–index regressions that quantify basis risk explicitly, and to extend the anomaly model with heavier tails for extreme drought.

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